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Comprehensive Analysis of Conductivity for Complex Acid-Etched Fractures: Findings from Ultra-Large-Scale Acid Fracturing Experiments

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Abstract

In carbonate reservoir stimulation, acid fracture conductivity is a paramount parameter for acid fracturing performance evaluation. Contemporary modeling paradigms exhibit limitations by concentrating primarily on simplified, geometrically ideal fractures at laboratory scales – a stark contrast to the intricate, heterogeneous fracture architectures encountered in actual field operations. More critically, the industry lacks methodologies for evaluating interconnected fracture systems' conductive properties. In light of this, a conductivity calculation model for complex acid-etched fractures through large-scale acid fracturing experiments was developed. Utilizing 2 m × 2 m × 1 m specimens, the research first obtains acid-etched fracture geometry and zero-closure-stress conductivity data. The methodology combines 3D laser scanning for fracture morphology and triaxial compression tests for post-etching mechanical properties, providing key inputs for model development and validation. The developed model combines an acid fracture closure mechanism based on linear elastic theory with continuity equations, enabling comprehensive conductivity calculations for complex fracture systems. Based on experimental results, the ultra-large-scale acid fracture network exhibits a fracture conductivity exceeding 10³ D·cm under zero closure stress, which is close to that calculated from the established model, verifying the accuracy of the model. The numerical simulation results indicate that the conductivity of complex fracture networks is significantly influenced by the characteristics of branch fractures. When the density of branch fractures is high, the connectivity between fractures improves, forming continuous flow channels that enhance overall fracture conductivity. The orientation of branch fractures, in conjunction with the dominant flow direction, also affects the conductivity of complex fracture systems. When the orientation of branch fractures aligns with the dominant flow direction, the conductivity of the complex fracture network improves significantly. Increasing the length of branch fractures not only enhances connectivity between fractures, but longer fractures also improve overall reservoir permeability by connecting a larger portion of the reservoir, thereby further enhancing the conductivity of the complex fracture network. As closure stress increases, the conductivity of branch

fractures decreases rapidly, and the overall conductivity of the system becomes increasingly dominated by the primary fracture, with the contribution of branch fractures diminishing. This newly developed acid-etched conductivity model for complex fracture systems is expected to provide fundamental tools for field treatment design.

Introduction

Carbonate reservoirs have long been regarded as a typical example of challenging formations for efficient development due to their complex pore structures and highly heterogeneous fracture systems. As one of the most effective stimulation techniques for carbonate reservoirs, acid fracturing involves injecting acid into hydraulically created fractures, where it reacts with minerals such as calcite and dolomite along the fracture walls (Nainwal et al., 2006; Long et al., 2010; Zhang and Wu, 2023). This reaction leads to uneven etching of the fracture surfaces, forming asperities of varying heights. After the treatment, the rough, acid-etched surfaces provide self-propping support under closure stress, preventing complete fracture closure and ultimately forming conductive acid-etched fractures. The non-uniform etching morphology of the fracture surfaces plays a decisive role in determining the post-fracturing conductivity (Ruffet et al., 1998; Malagon et al., 2006).

Currently, modeling studies on acid-etched fracture conductivity has formed a relatively comprehensive theoretical framework. The early N-K model (Nierode and Kruk, 1973) first incorporated parameters such as rock dissolution volume, closure stress, and embedment strength to establish an empirical regression-based conductivity model; however, it did not account for the influence of fracture surface roughness on conductivity. Subsequently, the bed-of-nails model (Gangi, 1978), Walsh model (1981), and void space model (Tsang and Witherspoon, 1981) introduced fracture roughness characteristics and stress sensitivity parameters, thereby enhancing the description of fracture deformation behavior and advancing fracture conductivity modeling from idealized geometries to representations more consistent with actual etched morphologies. The Gong model (Gong, 1997), based on 3D scanning reconstruction of fracture surfaces combined with experimental fitting, integrated rough particle distribution characteristics and stress correction coefficients into conductivity calculations, significantly improving model adaptability in representing non-uniform etching profiles. Since the 21st century, research focus has shifted from static characterization toward chemical–mechanical coupling mechanisms. Mou et al. (2009) introduced rock mineralogical heterogeneity and dissolution kinetics to establish a coupled relationship between microscale etching processes and macroscale fracture conductivity. The Mou-Deng model (Deng, 2010), utilizing finite element methods, established a multiscale coupling structure between roughness parameters and fluid flow behavior, thereby overcoming the limitations of the traditional parallel-plate fracture assumption. Kamali (2016) and Zhang et al. (2018) further incorporated the plastic yielding of asperities and the synergistic effects of normal and shear stresses during fracture closure, effectively enhancing the model's adaptability and predictive accuracy under complex in-situ stress conditions.

However, most acid-etched fracture conductivity models, developed based on simplified bi-wing fracture assumptions, have achieved widespread success in practical applications. Nevertheless, in fractured carbonate reservoirs, acid fracturing often creates fractures with complex geometries, such as multiple branches, irregular shapes, and uneven spatial distribution—features that significantly impact fracture conductivity. At present, there is a lack of systematic research and effective modeling approaches for quantifying the conductivity and understanding the response mechanisms of such complex fracture systems.

To address these issues, this study utilizes a 2 m × 2 m × 1 m ultra-large-scale physical acid fracturing experiment to obtain the characteristics of acid-etched fractures under realistic conditions, as well as their corresponding conductivity parameters. On this basis, a complex fracture conductivity calculation model incorporating the effects of closure stress was developed, and its accuracy and applicability were validated using experimental data. Furthermore, numerical simulations based on this model were performed

to investigate conductivity variations in complex fracture systems under different conditions, including branch fracture length, orientation, and density.

Experimental Method

The acid fracturing experiment includes hydraulic fracture initiation, acid injection, displacement, and post-acidizing fracture surface morphology scanning. Specifically, surface scans were performed on both sides of the fracture, and the scanning data were processed and combined to obtain the width distribution of the acid-etched fracture. Using these data along with the injection parameters recorded during the experiment, the corresponding fracture conductivity was calculated. These results provide experimental support to validate the subsequent complex fracture conductivity calculation model.

Experimental Materials and Equipment

The materials used in this ultra-large-scale acid fracturing experiment consist of rock samples, fracturing fluid, hydrochloric acid, and a gelling agent. The overall dimensions of the test block are $2\text{ m} \times 2\text{ m} \times 1\text{ m}$, with a carbonate outcrop (approximately $1.5\text{ m} \times 1.2\text{ m} \times 0.8\text{ m}$) embedded inside to simulate the reservoir rock matrix. To replicate the conditions of vertical well hydraulic fracturing, as shown in Fig. 1, the wellbore within the sample is oriented parallel to the vertical stress direction and is completed with a perforated casing. The outer diameter of the casing is 5.5 inches. Three perforation holes with a diameter of 1 cm are located 1 m below the specimen surface. These holes are distributed around the casing circumference at 120° phase intervals, with an axial spacing of 6 cm.

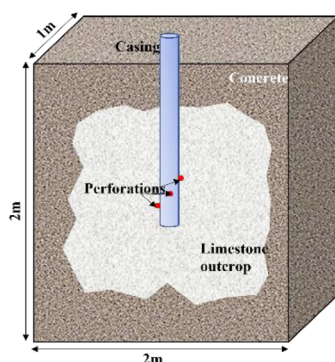


Figure 1—Schematic diagram of hydraulic fracturing

The applied three-dimensional stress conditions in the experiment were as follows: a minimum horizontal principal stress of 0 MPa, a maximum horizontal principal stress of 15 MPa, and a vertical stress of 25 MPa. Under these conditions, a fracturing fluid with a viscosity of $5\text{ mPa}\cdot\text{s}$ was injected at a rate of $0.3\text{ m}^3/\text{min}$ to fracture the rock sample. Following the completion of hydraulic fracturing, gelled acid was pumped into the fracture to simulate the etching behavior within the fractures during acid fracturing operations. Considering that the experimental setup could not be heated, the acid–rock reaction rate at ambient temperature was significantly lower than under actual subsurface high-temperature and high-pressure conditions. To enhance the reaction efficiency, a gelling agent was added at a concentration of 1%, maintaining the overall acid viscosity at approximately $5\text{ mPa}\cdot\text{s}$. This ensured sufficient flowability while accelerating the acid–rock reaction rate. The prepared acid solution was stored in sealed intermediate bulk container tanks, as shown in Fig. 2, to maintain concentration stability and prevent external contamination.



Figure 2—Tank for acid storage

The main equipment used in the experiment includes:

1. Large-scale rock sample stress loading system. This system is primarily used for applying stress to rock specimens in fracturing experiments. The loading chamber measures 2.5 meters in both length and width. It is equipped with a replaceable array of flat jacks, allowing flexible configuration of jack types and quantities to match the target stress conditions required for acid fracturing tests.
2. Lantong LTJ5330TYL200 fracturing truck. This unit is used for injecting fracturing fluids and acid solutions. It is equipped with a 3ZB105-875 plunger pump, with a maximum flow rate of 0.4 m³/min and a minimum flow rate of approximately 0.1 m³/min at idle. The pump is capable of delivering pressures up to 70 MPa.
3. Large-scale acid displacement pit. This facility is used for acid displacement and subsequent handling. The pit is divided into four functional zones: the specimen platform, spent acid neutralization pond, lime pool, and operation area. The specimen platform is positioned at the highest point of the pit, while the neutralization pond is at the lowest, with a height difference of 0.8 meters. This elevation gradient allows spent acid to flow naturally into the neutralization pond, preventing it from entering other areas. To protect groundwater from acid leakage, both the specimen platform and the bottom of the neutralization pond are lined with three layers of reinforced impermeable geomembrane. These structures and functional zones are illustrated in Fig. 3.
4. Auxiliary equipment. Other auxiliary equipment includes a crane, high-pressure pipelines, safety ropes, and mixing tanks.
5. After the experiment, a 3D profilometer is used to scan and measure the fracture surface in order to obtain the morphology of the acid-etched fractures.

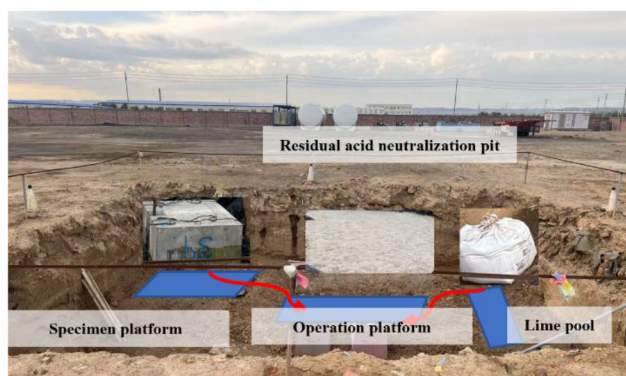


Figure 3—Large acid displacement pit



Figure 4—Crane

Experimental Procedure

Hydraulic Fracturing for Fracture Creation. Fractures were first generated through hydraulic fracturing, and the fracturing process was terminated once the fractures penetrated the rock specimen.

Hoisting of the rock specimen and equipment assembly. After hydraulic fracturing, the rock specimen was transferred onto the specimen platform within the large acid displacement pit using a heavy-duty crane. Wooden planks were employed to stabilize and support the specimen, ensuring that residual acid could flow smoothly into the neutralization tank during the acid injection process. Subsequently, high-pressure pipelines were used to connect the fracturing truck and the mixing tank to the rock specimen. To ensure operational safety, safety ropes were properly installed as required. Meanwhile, hydrated lime was hoisted into the lime pool for standby use. Finally, all pipeline connections were carefully inspected to confirm the absence of any leakage, thereby ensuring the safety and reliability of the experimental operations.

Acid Solution Preparation. A crane was initially employed to lift the IBC acid container to an appropriate height relative to the mixing tank, after which the acid was slowly introduced into the tank. The stirrer was subsequently activated, and the gelling agent was gradually added while maintaining continuous stirring for one hour to ensure sufficient mixing between the acid and the gelling agent. If undissolved gelling agent was observed after stirring, a mesh filter was used to remove the remaining residues, thereby improving the purity and effectiveness of the reaction. Ultimately, a total of two tons of acid solution was successfully prepared.

Acid Displacement. After the acid solution was fully prepared, the fracturing truck was activated with a preset injection rate of $0.1 \text{ m}^3/\text{min}$. The acid was then injected into the rock specimen along the wellbore, with a total injection duration of 18 minutes. During the acid injection process, real-time monitoring of pump pressure was required to provide reliable data for subsequent conductivity calculations. Upon completion of acid injection, clean water was immediately injected to displace residual acid in the equipment pipelines and within the fracture. Subsequently, one ton of hydrated lime was slowly added to the residual acid neutralization tank to carry out acid neutralization, ensuring complete reaction of the spent acid. Finally, environmentally safe disposal procedures were implemented to ensure that the entire treatment process met environmental safety standards. The overall procedure is illustrated in Fig. 5.

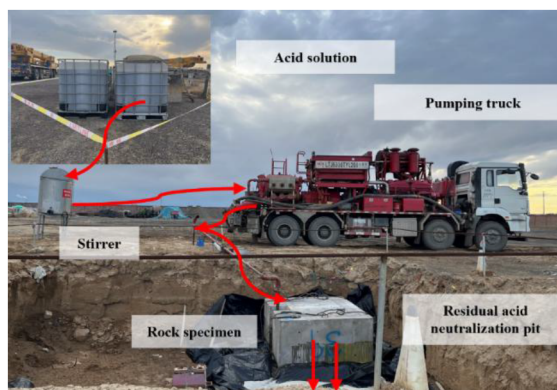


Figure 5—Acid displacement equipment and its layout

Sectioning of the Specimen and Topographical Scanning of the Fracture Surface. After the acid injection was completed, the specimen was first hoisted and removed, and any residual acid on its surface was thoroughly cleaned. Subsequently, the specimen was lifted again using a crane, and a hydraulic spreader was employed to split the rock along the fracture plane. The exposed fracture surfaces were then subjected to topographical scanning to obtain detailed surface morphology data. These data were used to reconstruct the three-dimensional geometry of the fractures and to calculate the width distribution of the acid-etched fractures based on the scanned profiles.

Experimental Results and Analysis

Complex Fracture Morphology. After the fracturing experiment, a complex fracture system comprising a primary fracture and secondary fractures was formed, exhibiting an overall inverted T-shape. The primary fracture propagated along the direction of the maximum horizontal principal stress and nearly penetrated the entire rock specimen, with localized bending and branching, as indicated by the red dashed lines in Fig. 6. The secondary fractures intersect the primary fracture obliquely, as illustrated by the blue dashed lines. The primary fracture has a relatively large aperture, reaching up to 2 mm at its widest point, whereas the secondary fractures are much narrower and barely discernible to the naked eye.



Figure 6—Complex fracture morphology after fracturing

Reconstruction of Acid-Etched Fracture Surfaces and Dissolution Characteristics. The 3D morphology scanning data of the fracture contains X, Y, and Z coordinate information. By applying appropriate coordinate transformations and calculations to the scanned morphology data of the upper and lower fracture surfaces, the width distribution of the acid-etched fracture can be obtained. For a given fracture, the

morphology data of both surfaces are first processed using a unified coordinate transformation, taking the midpoint of the bottom surface of the specimen on the same side as both fracture surfaces (as shown in Fig. 8) as the reference origin. Interpolation is then applied to ensure one-to-one correspondence of X and Y coordinates between the upper and lower fracture surfaces, achieving geometric alignment of the two surfaces, as illustrated in Fig. 7. Based on the aligned coordinates, the Z-coordinate data of the upper and lower surfaces are added to obtain the height distribution of the composite dissolution surface, denoted as $Z_{composite}$. The dissolution fracture width distribution is then calculated using $W_i = H - Z_{composite}$. Finally, the resulting fracture width distribution is used as the initial condition for fracture deformation calculations.

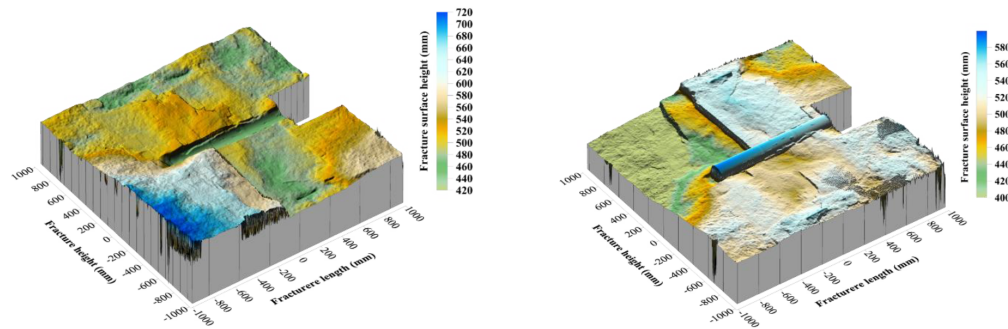


Figure 7—Reconstructed etched fracture surface of the main fracture

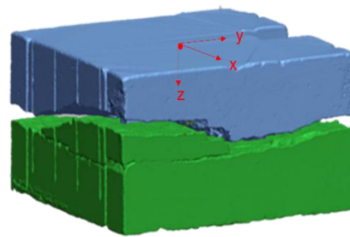


Figure 8—Schematic Diagram of the Reference Origin for Fracture Surface Alignment

The processed fracture width distribution is shown in Fig. 9. It can be observed that the acid-etched fracture width is relatively large in the lower-left region, primarily because this area is directly aligned with the perforation. Acid predominantly exits through this perforation. In the near-wellbore region, the acid flow velocity and concentration are higher, and the contact time between acid and rock is longer, leading to more significant dissolution and pronounced grooving. In the middle section of the fracture, the acid flow becomes more dispersed and diluted, resulting in less overall dissolution and a reduced degree of differential dissolution compared to the perforation zone. The remaining fracture regions, located farther from and opposite to the perforation, receive less acid, leading to limited etching and correspondingly smaller acid-etched fracture widths.

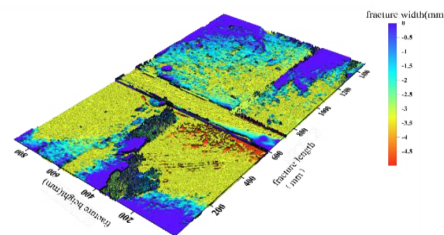


Figure 9—Main fracture aperture profile

Fracture Conductivity. According to the fracture conductivity calculation formula and the corresponding injection parameters, the fracture conductivity obtained from this experiment was calculated to be $1.32 \times 10^3 \text{ D} \cdot \text{cm}$. This result will serve as a validation basis for the complex fracture conductivity calculation model developed in the subsequent study.

$$q = \frac{kwh\Delta p}{\mu L} \quad (1)$$

Rock mechanical parameters. Three sets of standard-size core samples were drilled from the acid-etched fracture surfaces in the above-mentioned ultra-large-scale acid fracturing experiment. These cores were used for triaxial compression tests (Zhou et al., 2020) to obtain the rock mechanical parameters of the acid-etched fracture surfaces under a confining pressure of 45 MPa. The measured mechanical parameters are shown in Table 1. The results indicate that, at the experimental scale, the Young's modulus of cores containing acid-etched fracture surfaces ranges from 6 GPa to 10 GPa, and the failure strength ranges from 85 MPa to 97 MPa. Specifically, the failure loads for cores P1, P2, and P3 were 21,597 N, 25,476 N, and 20,854 N, corresponding to failure pressures of 87.6 MPa, 96.2 MPa, and 86.3 MPa, respectively (failure pressure = differential stress + confining pressure). The corresponding Young's moduli were 7.62 GPa, 9.13 GPa, and 6.29 GPa. These results demonstrate that acid dissolution on the fracture surface has a significant degrading effect on the Young's modulus. Since Young's modulus characterizes the resistance to deformation of the fracture surface, a lower modulus indicates more severe deformation of the acid-etched surface, greater loss of fracture aperture, and faster decline in conductivity with increasing closure stress. These data are used for fracture surface deformation and fracture width calculations.

Table 1—Rock Mechanical Parameters of Acid-Etched Fracture Surfaces

Sample ID	Confining pressure (MPa)	Length (mm)	Diameter (mm)	Load (N)	Differential stress (MPa)	Young's modulus (GPa)
P1	45	47.56	24.96	21597	42.6	7.62
P2	45	45.62	24.96	25476	51.2	9.13
P3	45	46.16	24.96	20854	41.3	6.29



Figure 10—Fracture surface of the fractured samples after triaxial compression test

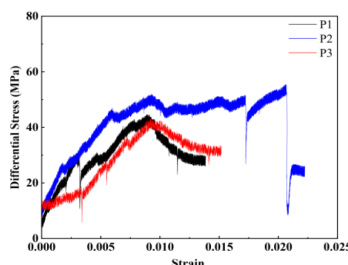


Figure 11—Stress-strain curve

Complex Fracture Conductivity Calculation Model

For carbonate reservoirs with complex fracture geometries formed by acid fracturing, this study establishes a computational model for evaluating the conductivity of complex fractures. The model development includes the following key steps: First, it is recognized that the non-uniform acid–rock reaction along the fracture surfaces leads to the formation of rough structures with significant spatial variations in height. Under closure stress, asperities on the opposing fracture surfaces come into contact (Brown and Scholz, 1985; Greenwood and Williamson, 1966; Greenwood and Tripp, 1967) and undergo elastic deformation (Jackson et al., 2006), resulting in changes in the aperture of the acid-etched fractures. To quantitatively describe this process, linear elasticity theory is used to calculate the deformation of the rough micro-asperities, thereby obtaining the fracture aperture distribution under closure stress. By combining the steady-state flow equation within the fracture with the deformed fracture aperture as the input parameter, the conductivity (Wu and Sharma, 2017) of a single fracture under closure stress is calculated, as shown in Figs. 12 and 13. To improve computational efficiency, the three-dimensional fracture system is simplified into a two-dimensional fracture network through geometric projection, neglecting flow in the vertical (height) direction of the fractures. Based on this simplification, the conductivity of each fracture under different closure stresses is converted into equivalent permeability values, enabling the construction of a permeability field for the complex fracture system, as illustrated in Fig. 14. The continuity equation is then used again to solve for the pressure and flow distribution within the fractured domain. Finally, the effective conductivity of the complex fracture network is calculated.

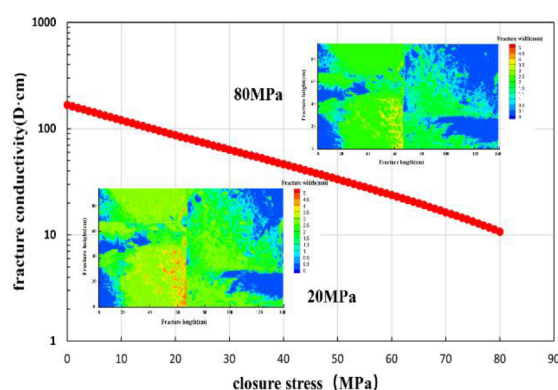


Figure 12—Main fracture conductivity

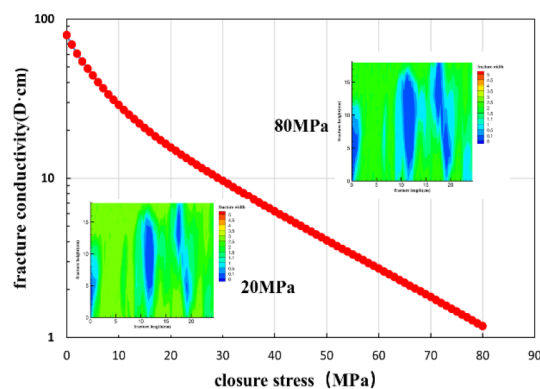


Figure 13—Branch fracture conductivity

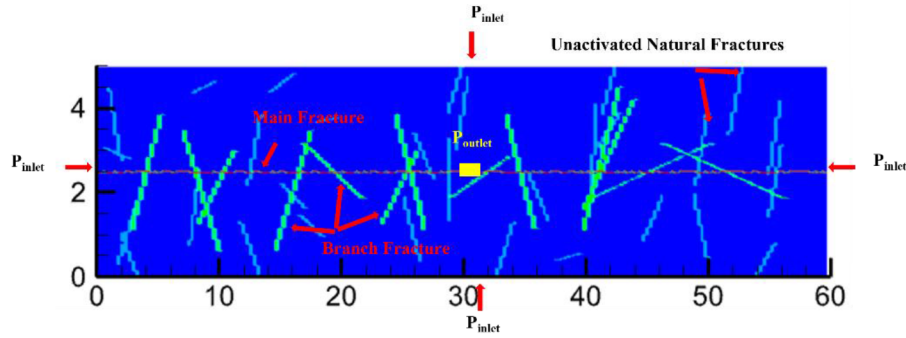


Figure 14—Complex fracture model

Fracture Closure Model

Assuming that the deformation of the fracture under closure stress is purely elastic, the classical Hooke's law is employed to quantitatively analyze the deformation of rough micro-asperities on the fracture surfaces. The contact ratio σ is introduced and defined as the ratio of the actual contact area A_c to the nominal area A , as shown in Equation (2).

$$\alpha = \frac{A_c}{A} \quad (2)$$

When a stress σ is applied to the fracture surface, the rough micro-asperities on the surface bear a stress σ' . The relationship between σ and σ' is expressed as follows:

$$\sigma' = \frac{\sigma}{\alpha} \quad (3)$$

Let the deformation of the rough micro-asperities on the fracture surface be denoted as Δh , and the initial height as h . According to Hooke's law, the following relationship can be obtained:

$$\sigma' = E\varepsilon = E \frac{\Delta h}{h} \quad (4)$$

where E is the Young's modulus of the material, and ε is the strain.

By combining Equations (4) and (5), we obtain:

$$\Delta h = \frac{\sigma h}{\alpha E} \quad (5)$$

For the rough micro-asperities in contact on the opposing fracture surfaces, assuming the initial fracture aperture distribution is denoted as W_i and the deformation under a given closure stress is Δh , the deformed aperture distribution can be expressed as:

$$W = W_i - \Delta h \quad (6)$$

Single-Fracture Conductivity Calculation Model

The model is developed based on the fundamental assumption of two-dimensional steady-state linear flow, with fluid flow governed by the classical Darcy's law. The flow rate along the fracture length and height is calculated based on the local cubic law as follows:

$$q_x = \frac{W^3 H_x}{12\mu} \frac{\partial p}{\partial x} \quad (7)$$

$$q_y = \frac{W^3 H_y}{12\mu} \frac{\partial p}{\partial y} \quad (8)$$

Here, q_x and q_y represent the flow rates in the fracture length and height directions, respectively; h is the grid block size, μ is the fluid viscosity, and p is the fluid pressure. The fracture closure model provides the aperture W for each grid block under the given closure stress. Since the aperture is much smaller than

the fracture length and height, flow perpendicular to the fracture surfaces can be neglected. The continuity equation for two-dimensional flow is expressed as:

$$\frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} = 0 \quad (9)$$

Substituting Equations (7) and (8) into Equation (9) yields the following partial differential equation:

$$\frac{\partial}{\partial x} \left(W^3 \frac{\partial P}{\partial x} \right) + \frac{\partial}{\partial y} \left(W^3 \frac{\partial P}{\partial y} \right) = 0 \quad (10)$$

It is important to note that the grid cells in this model are square elements. Regarding boundary condition settings, the left boundary is defined as the inlet pressure boundary, while the right boundary is defined as the outlet pressure boundary. The upper and lower boundaries are set as no-flow (closed) boundaries. The equation is discretized and solved using the finite difference method. Based on the fracture aperture distribution under different closure stresses, the pressure field within the fracture can be calculated. The total flow rate within the fracture is then determined from the pressure distribution along the inlet boundary. Finally, the conductivity of a single acid-etched fracture is obtained according to Equation (11).

$$C_f = K_f W = \frac{q \mu L}{H \Delta P} \quad (11)$$

Where, L is the fracture length and H is the fracture height. Based on the definition of flow conductivity, the concept of equivalent fracture width is introduced to derive the equivalent permeability of the fracture.

$$K_f = \frac{C_f}{W_f} \quad (12)$$

Where, W_f is the equivalent fracture aperture and C_f is the fracture conductivity.

Complex Fracture Conductivity Calculation Model

Assuming that the branch fractures and the main fracture have the same height, the three-dimensional fracture network is projected onto a two-dimensional plane, simplifying the flow within the complex fracture network to two-dimensional flow in the horizontal plane. The equivalent permeability of each fracture under different closure stresses, obtained from the single-fracture conductivity model, is assigned to the corresponding fractures. Meanwhile, the grid cells outside the fractures are assigned the matrix permeability value. The continuity Equation (13) is then used again to solve for the pressure distribution under steady-state flow conditions within the complex fracture system. To reflect realistic flow conditions, constant-pressure inflow boundaries are applied on all four sides (top, bottom, left, and right) of the complex fracture region, and a constant-pressure outflow boundary is set at the center of the main fracture. Finally, the overall conductivity of the complex fracture network under different closure stresses is calculated using the fracture conductivity Equation (12).

$$\frac{\partial}{\partial x} \left(k \frac{\partial P}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial P}{\partial y} \right) = 0 \quad (13)$$

Model Validation

Based on the complex fracture morphology and fracture aperture distribution characteristics obtained from the ultra-large-scale acid fracturing experiment, the above-described complex fracture conductivity model was used to conduct numerical simulations, as shown in Fig. 15. The simulation results show that the computed fracture conductivity of the complex fracture network is 1.57×10^3 D·cm, which is close to the experimentally measured conductivity 1.32×10^3 D·cm, with only a small discrepancy between the two. This confirms the accuracy and applicability of the proposed complex fracture conductivity model.

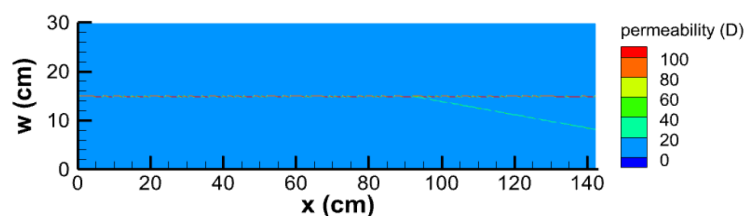


Figure 15—Projected fracture geometry from ultra-large-scale acid fracturing experiment

Variation Pattern of Complex Fracture Conductivity

In acid-fractured complex fracture systems, the fracture network typically consists of a main fracture and multiple surrounding branch fractures. A main fracture of specified length is first generated in the center of the study area, and its initial acid-etched aperture distribution is obtained using the acid fracturing model (Mou, 2009). Then, the spatial distribution of natural fractures in the horizontal plane is generated using the Sequential Gaussian Simulation (SGS) method. The natural fractures that are connected to the main fracture are defined as branch fractures. This setup is reasonable because, during acid fracturing, the acid can flow along the main fracture and enter the intersecting natural fractures, where acid–rock reactions occur, thereby activating the natural fractures as secondary fractures. The initial acid-etched aperture distribution of the secondary fractures is assigned based on the statistical characteristics observed in laboratory experiments. This study focuses on analyzing the impact of branch fractures on fracture conductivity. The constructed fracture network is then mapped onto the computational grid used in the complex fracture conductivity model, providing the basis for subsequent conductivity analysis.

The characteristics of branch fractures in a complex fracture network—such as length, density, and orientation—largely determine the geometric complexity and spatial connectivity of the fracture system, and thus influence its overall conductivity. To investigate the impact of branch fracture geometry on the conductivity of complex fracture systems, this study adopts a controlled variable approach. While keeping other parameters constant, multiple levels are assigned to the target parameters (length, density, and orientation), and a series of branch fracture networks with varying features are constructed. These networks are then input into the complex fracture conductivity model to systematically analyze the specific influence of each parameter on the conductivity of the complex fracture system. The simulation parameters are presented in Table 2.

Table 2—Simulation parameters

Study area:60 m×5 m		Young's modulus:7.5 GPa	Inlet pressure:10 MPa Outlet pressure:0 MPa	
Branch Fracture Density:0.1-0.3 fractures/m		Branch Fracture Orientation :0°-90°	Branch Fracture Length:1-7m	
Closure stress (MPa)	Main fracture conductivity (D·cm)	Main fracture permeability (D)	Branch Fracture conductivity (D·cm)	Branch Fracture permeability (D)
0	166.8447369	133.4757895	79.49741666	63.59793333
10	120.3276316	96.26210526	28.91141666	23.12913333
20	86.66513158	69.33210526	17.40283334	12.51413333
30	63.2868421	69.33210526	9.655166666	7.724133333
40	46.24671053	36.99736842	6.210583334	4.968466667
50	33.36973684	26.69578947	4.086083334	3.268866667
60	23.69736843	18.95789474	2.941833334	2.168533333
70	16.33881579	13.07105263	1.796583334	1.437266667
80	10.70986842	8.567894737	1.181083334	0.944866667

Density of Branch Fractures

Based on the fracture network construction method described above, three complex fracture networks were generated with branch fracture densities of 0.1 fractures/m, 0.2 fractures/m, and 0.3 fractures/m, respectively, while keeping the branch fracture length (5–7 m) and orientation (0–60°) constant. As shown in Fig. 16, the simulation results of complex fracture system conductivity under different branch fracture densities indicate that changes in branch fracture density have a significant impact on the connectivity of the fracture network. As the density of branch fractures increases, the number of fractures increases accordingly, which raises the probability of fracture intersections. This enhances the connectivity between fractures and improves the overall connectivity of the fracture network structure. During fluid flow, higher connectivity facilitates the formation of continuous and stable flow paths within the fracture system, thereby effectively enhancing the overall conductivity of the complex fracture network. In contrast, when the branch fracture density is low, the number of fractures decreases, significantly reducing the chances of intersection and connection among fractures. This results in a limited number of flow paths, restricted network connectivity, and impaired fluid transport throughout the system. Therefore, branch fracture density is one of the key factors influencing the conductivity of complex fracture systems. Further analysis reveals that the influence of branch fracture density on the conductivity of complex fracture systems varies significantly under different closure stress conditions. Under low closure stress, both the main fracture and branch fractures can maintain a certain level of conductivity, and together they contribute to the overall system conductivity. As a result, the differences in conductivity among cases with varying branch fracture densities are more pronounced. However, because the conductivity of branch fractures decays more rapidly than that of the main fracture with increasing closure stress, their contribution to flow becomes significantly weaker under high closure stress. In this case, the system's conductivity is primarily governed by the main fracture, leading to a reduced difference in conductivity across different branch density scenarios. In summary, appropriately increasing the density of branch fractures—especially under low closure stress—can improve fracture network connectivity, promote the formation of effective flow channels, and thereby enhance the overall conductivity of complex fracture systems.

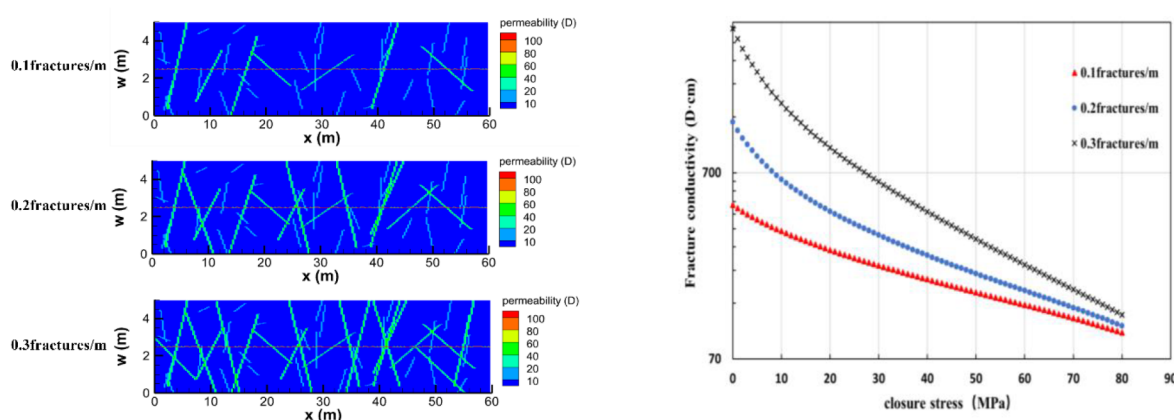


Figure 16—Simulation results of complex fracture conductivity under different branch fracture densities

Orientation Angle of Branch Fractures

Based on the fracture network construction method described above, three complex fracture networks were generated with branch fracture orientations of 0–30°, 30–60°, and 60–90°, respectively, while keeping the branch fracture length (5–7 m) and density (0.2 fractures/m) constant. It is important to note that the term "orientation" here refers to the angle between the branch fractures and the main fracture. As shown in Fig. 17, analysis of the complex fracture system conductivity under different branch fracture orientation conditions reveals that, under low closure stress, the overall conductivity of the fracture system gradually increases

with increasing branch fracture orientation angle. This phenomenon is closely related to the characteristics of the simulation domain. The flow model in this study is constructed as a narrow rectangular region with dimensions of 60 m $\times$ 5 m, with constant-pressure inlet boundaries applied on all four sides and a fixed-pressure outlet at the center. Because the model width (short axis) is significantly smaller than its length (long axis), the pressure gradient in the width direction is much higher than in the length direction, making it easier for the primary flow paths to develop along the width. Against this background, when the branch fracture orientation falls within the 60°–90° range, their alignment is more consistent with the dominant flow direction. This enables more effective coupling and connection with the main flow paths, significantly reducing local flow resistance and thereby enhancing the overall conductivity of the fracture system. Further analysis indicates that, similar to the effect of branch fracture density, the influence of branch fracture orientation on the system conductivity exhibits a consistent trend under different closure stress conditions. In summary, optimizing the orientation of branch fractures—particularly under low closure stress—to enhance their alignment with the dominant flow direction can significantly improve the overall conductivity of complex fracture systems.

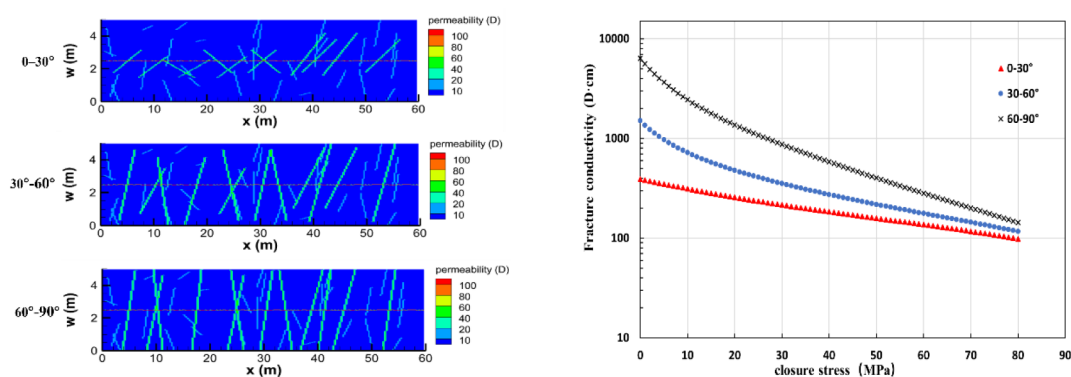


Figure 17—Simulation results of complex fracture conductivity under different branch fracture orientation angles

It is worth noting that in actual formations, the combined effects of in-situ stress fields, lithological heterogeneity, and fracture network architecture lead to a pronounced anisotropy in the pressure gradient. This directional pressure distribution drives fluid flow preferentially along specific paths within the natural fracture system, forming dominant flow channels. Therefore, the simulation results of this study provide a theoretical basis and valuable reference for analyzing the directional evolution of flow conductivity in fracture systems under real reservoir conditions.

Length of Branch Fractures

Based on the fracture network construction method described above, three complex fracture networks were generated with branch fracture lengths of 1–3 m, 3–5 m, and 5–7 m, respectively, while keeping the branch fracture density (0.2 fractures/m) and orientation (0–60°) constant. As shown in Fig. 18, analysis of the flow conductivity of complex fracture networks under different branch fracture lengths indicates that branch length has a significant impact on the degree of intersection among fractures, thereby affecting both the number and spatial distribution of fracture intersections within the network. When the branch fractures are longer, intersections between fractures are more likely to occur, leading to the formation of a more complex and better-connected fracture network. This enhanced connectivity facilitates the development of flow channels and contributes to improved overall flow conductivity of the system. Therefore, as the length of the branch fractures increases, the conductivity of the complex fracture network tends to improve accordingly. In addition, longer branch fractures can span a greater number of grid cells, expanding the spatial coverage of flow pathways and enabling connectivity to larger-scale reservoir regions, which further enhances the effective flow capacity of the fracture network. In contrast, shorter branch fractures typically

provide only localized connectivity and are less capable of forming large-scale continuous flow paths, thus offering limited contribution to improving the overall flow conductivity of the system. Further analysis shows that, similar to the effect of branch fracture density, the influence of branch fracture length on system conductivity exhibits a consistent trend under different closure stress conditions. Therefore, appropriately increasing the length of branch fractures—especially under low closure stress—can significantly enhance the overall conductivity of complex fracture systems.

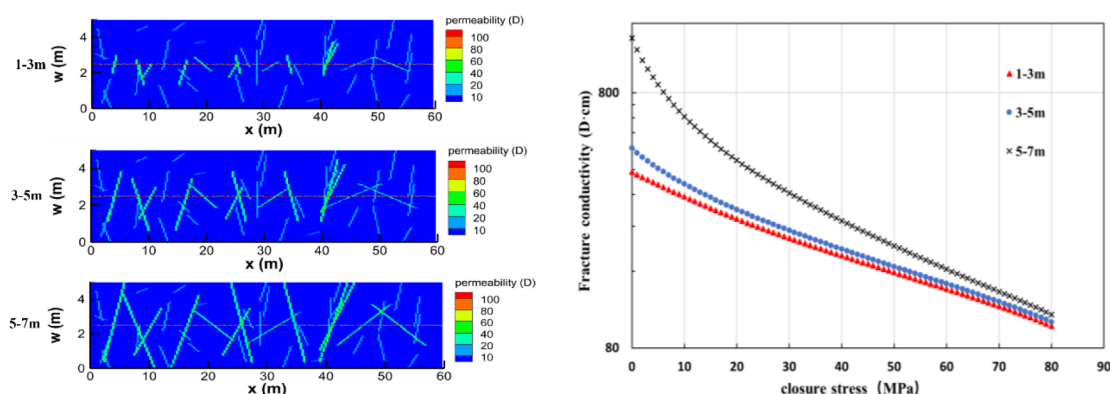


Figure 18—Simulation Results of Conductivity in Complex Fracture Networks Under Different Branch Fracture Lengths

Conclusion

This study is grounded in ultra-large-scale acid fracturing experiments, through which the morphology of fractures, mechanical properties of the etched fracture surface, and fracture conductivity were obtained. Based on these data, a complex fracture conductivity calculation model was established, and its accuracy and applicability were validated using measured experimental results. Furthermore, the study systematically examined the influence of key parameters of the fracture network—specifically, branch fracture density, orientation angle, and length—on fracture conductivity of the complex fracture system. Integrating experimental observations with numerical simulation analysis, the following conclusions were drawn:

1. The acid-etched fracture conductivity obtained from the ultra-large-scale acid fracturing experiment was measured under zero closure stress conditions; therefore, the measured conductivity is relatively high, exceeding 10^3 D·cm.
2. As the branch fracture density increases, the number of fractures rises and the probability of intersections becomes higher, which enhances connectivity between fractures and promotes the formation of continuous and stable flow channels, thereby improving the overall conductivity of the complex fracture system.
3. The effect of branch fracture orientation on fracture conductivity is strongly influenced by its alignment with the dominant flow direction. When branch fractures are more aligned with the primary flow paths, local flow resistance decreases significantly, thereby improving the overall conductivity of the fracture system.
4. As the length of branch fractures increases, they are more likely to intersect with other fractures in space, thereby improving the connectivity of the fracture system. In addition, longer fractures can penetrate more reservoir units, expanding the spatial coverage of flow pathways and enhancing the overall permeability of the reservoir. These two factors act synergistically, resulting in a substantial increase in the overall conductivity of the complex fracture network with increasing branch fracture length.
5. At low closure stress, the distribution of branch fractures significantly influences the conductivity of complex fracture systems. As closure stress increases, the conductivity of branch fractures decreases

rapidly, and the overall conductivity of the system becomes increasingly dominated by the primary fracture, with the contribution of branch fractures diminishing.

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